

Yongan (Luke) Zhang

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RESEARCH INTEREST

- LLM agents for hardware design automation
- Efficient ML systems and software-hardware co-design

EDUCATION

- **Georgia Tech** Atlanta, GA, USA
PhD, Computer Science, Advisor: Prof. Yingyan Lin Aug. 2023 – May. 2025
- **Rice University** Houston, TX, USA
MS, Electrical and Computer Engineering, Advisor: Prof. Yingyan Lin Jan. 2020 – May 2023
- **Rice University** Houston, TX, USA
BS, Electrical and Computer Engineering Aug. 2015 – May 2019

EXPERIENCES

- **Research Engineer, Meta** Manager: Andrii Gushchik
Efficient ML Virtual Telepresence May 2025 - Present
 - Develop compiler- and system-level optimizations for efficient on-device ML inference, supporting quantization, pruning, and adaptive runtime techniques across heterogeneous mobile accelerators for virtual telepresence applications.
- **Research Intern, Meta** Mentor: Dr. Yuecheng Li
Adaptive Once-for-all model compression May 2024 - Jan 2025
 - Designed a once-for-all model compression framework for audio-driven real-time human avatar generation, enabling efficient design space exploration and flexible deployment.
 - Investigated runtime-adaptive model compression as a domain-specific acceleration opportunity, including micro-architectural support for dynamic model shapes and sparsity.
- **Research Intern, Meta** Mentor: Dr. Yuecheng Li
Reconfigurable hardware acceleration for VR mobile telepresence pipeline May 2022 - Dec 2022
 - Designed a runtime-reconfigurable architecture for improved hardware resource efficiency.
 - Developed fine-grained operation scheduling for model-to-hardware mapping.
 - Built RTL-verified performance modeling for flexible design space exploration.
 - Implemented an automation flow that auto-generates the micro-architecture design and scheduling.
- **Ph.D. Intern, PNNL** Mentor: Dr. Ang Li
Multi-FPGA acceleration for scalable Graph Neural Networks implementation Jan 2022 – May 2022
 - Designed a distributed multi-FPGA architecture for scalable GNN inference.
 - Took the design from architecture specification to board-level deployment.

PUBLICATIONS

LLM Agents for Hardware Design Automation

1. **Y. Zhang**, Z. Yu, Y. Fu, C. Wan, Y. Lin, “MG-Verilog: Multi-grained Dataset Towards Enhanced LLM-assisted Verilog Generation”, *1st IEEE International Workshop on LLM-Aided Design (LAD)*, 2024, *Best Paper*.
2. **Y. Zhang**, Y. Fu, Z. Yu, K. Zhao, C. Wan, C. Li, Y. Lin, “INVITED: Data4AIGChip: An Automated Data Generation and Validation Flow for LLM-assisted Hardware Design”, *The 58th Design Automation Conference (DAC)*, 2024.
3. Y. Fu*, **Y. Zhang***, Z. Yu*, S. Li, Z. Ye, C. Li, C. Wan, Y. Lin, “GPT4AIGChip: Towards Next-Generation AI Accelerator Design Automation via Large Language Models”, *ACM/IEEE International Conference On Computer Aided Design (ICCAD)*, 2023.

Automated Design Frameworks for ML Accelerators

1. **Y. Zhang**, X. Zhang, P. Xu, Y. Zhao, C. Hao, D. Chen, Y. Lin, “AutoAI2C: An Automated Hardware Generator for DNN Acceleration On Both FPGA and ASIC”, *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, 2024.
2. Y. Zhao, C. Li, Y. Wang, P. Xu, **Y. Zhang**, Y. Lin, “DNN-Chip Predictor: A Multi-grained Graph-based Performance Simulator for DNN Accelerators”, *International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2020.
3. P. Xu, Y. Zhao, C. Hao, X. Zhang, Z. Guan, **Y. Zhang**, Y. Wang, D. Chen, Y. Lin, “AutoDNNchip: An Automated DNN Chip Predictor and Builder for Both FPGAs and ASICs”, *ACM/SIGDA International Symposium on Field-Programmable Gate Arrays (FPGA)*, 2020.

Algorithm–Accelerator Co-Search and Co-Design

1. **Y. Zhang**, H. You, Y. Fu, T. Geng, A. Li, Y. Lin, “G-CoS: GNN-Accelerator Co-Search Towards Both Better Accuracy and Efficiency”, *IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, 2021.
2. **Y. Zhang**, Y. Fu, W. Jiang, C. Li, H. You, M. Li, V. Chandra, Y. Lin, “DIAN: Differentiable Accelerator-Network Co-Search Towards Maximal DNN Efficiency”, *ACM/IEEE International Symposium on Low Power Electronics and Design (ISLPED)*, 2021.
3. **Y. Zhang**, A. Banta, Y. Fu, M. John, A. Post, M. Razavi, J. Cavallaro, B. Aazhang, Y. Lin, “RT-RCG: Neural Network and Accelerator Search Towards Effective and Real-time ECG Reconstruction from Intracardiac Electrograms”, *The ACM Journal on Emerging Technologies in Computing Systems (JETC)*, 2021.
4. Y. Fu, **Y. Zhang**, H. You, Y. Lin, “Auto-NBA: Efficient and Effective Search Over The Joint Space of Networks, Bitwidths, and Accelerators”, *The International Conference on Machine Learning (ICML)*, 2021.
5. Y. Fu, **Y. Zhang**, C. Li, Z. Yu, Y. Lin, “A3C-S: Automated Agent Accelerator Co-Search towards Efficient Deep Reinforcement Learning”, *The 58th Design Automation Conference (DAC)*, 2021.
6. Y. Fu, Z. Yu, **Y. Zhang**, Y. Jiang, C. Li, Y. Liang, M. Jiang, Z. Wang, Y. Lin, “InstantNet: Automated Generation and Deployment of Instantaneously Switchable-Precision Networks”, *The 58th Design Automation Conference (DAC)*, 2021.
7. M. Li, Z. Yu, **Y. Zhang**, Y. Fu, Y. Lin, “O-HAS: Optical Hardware Accelerator Search for Boosting Both Acceleration Performance and Development Speed”, *IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, 2021.
8. C. Li, Z. Yu, Y. Fu, **Y. Zhang**, Y. Zhao, H. You, Q. Yu, Y. Wang, Y. Lin, “HW-NAS-Bench: Hardware-Aware Neural Architecture Search Benchmark”, *The International Conference on Learning Representations (ICLR)*, 2021.

Domain-Specific Accelerators and Efficient ML Systems

1. **Y. Zhang**, Y. Li, S. Sarwar, H. Sumbul, Y. Fu, H. You, C. Wan, Y. Lin, “Re-CATA: Real-Time and Flexible Accelerator Design Framework for On-device Codec Avatars”, *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, 2025.
2. Z. Ye, Y. Fu, J. Zhang, L. Li, **Y. Zhang**, S. Li, C. Wan, C. Li, S. Prathipati, Y. Lin, “Gaussian Blending Unit: An Edge GPU Plug-in for Real-Time Gaussian-Based Rendering in AR/VR”, *IEEE International Symposium on High Performance Computer Architecture (HPCA)*, 2025.
3. Y. Fu, Z. Yu, J. Li, J. Qian, **Y. Zhang**, X. Yuan, D. Shi, R. Yakunin, Y. Lin, “AmoebaLLM: Constructing Any-Shape Large Language Models for Efficient and Instant Deployment”, *The 38th Conference on Neural Information Processing Systems (NeurIPS)*, 2024.
4. Y. Zhao, **Y. Zhang**, Y. Fu, X. Ouyang, C. Wan, S. Wu, A. Banta, M. John, A. Post, M. Razavi, J. Cavallaro, B. Aazhang, Y. Lin, “e-G2C: A 0.14-to-8.31 uJ/Inference NN-based Processor with Continuous On-chip Adaptation for Anomaly Detection and ECG Conversion from EGM”, *IEEE Symposium on VLSI Technology and Circuits (VLSI)*, 2022.
5. H. You, T. Geng, **Y. Zhang**, A. Li, Y. Lin, “GCoD: Graph Convolutional Network Acceleration via Dedicated Algorithm and Accelerator Co-Design”, *IEEE International Symposium on High-Performance Computer Architecture (HPCA)*, 2022.
6. H. You, Y. Zhao, Z. Yu, C. Wang, Y. Fu, J. Yuan, S. Wu, S. Zhang, **Y. Zhang**, C. Li, V. Boominathan, A. Veeraraghavan, Z. Li, Y. Lin, “EyeCoD: Eye Tracking System Acceleration via FlatCam-Based Algorithm and Accelerator Co-Design”, *IEEE/ACM International Symposium on Computer Architecture (ISCA)*, 2022.
7. T. Geng, C. Wu, **Y. Zhang**, C. Tang, C. Xie, H. You, M. Herbordt, Y. Lin, A. Li, “I-GCN: A Graph Convolutional Network Accelerator with Runtime Locality Enhancement through Islandization”, *IEEE/ACM International Symposium on Microarchitecture (MICRO)*, 2021.
8. H. You, X. Chen, **Y. Zhang**, C. Li, S. Li, Z. Liu, Z. Wang, Y. Lin, “ShiftAddNet: A Hardware-Inspired Deep Network”, *Conference on Neural Information Processing Systems (NeurIPS)*, 2020.

AWARDS

- IEEE LAD Best Paper 2024
- 1st & 2nd Place University Demonstration at DAC 2022 & 2023
- Distinction in Research and Creative Work 2019